PLASTERING:

Description:

$\frac{1}{2}$" thick cement plaster of ratio 1:6 is used.

Calculation for volume:

① LONG WALL ①:

length = 7'3" = 7.25'
height = 10'

covered area = 7.25 x 10 = 72.5 sq. ft

② LONG WALL ②:

⇒ length (~~Rebidé~~ in ko in) = 7.25'
height = 10'

covered area = 72.5 sq.ft

⇒ length (out to out) = 7'3" + 4.5" - 4.5"
= 7'3" = 7.25'
height = 10' ⇒ Area of wall = 72.5'
Door area = 2.5' x 7' = 17.5 sq ft

covered area = 72.5 - 17.5 = 55 sq ft

③ SHORT WALL 1:

3551.83

Length = 5'
height = 10'

covered area = 50 sq ft

④ SHORT WALL 2:

Length (Bath side) = 5'
height = 10'

covered area = 50 sq ft

(4) SHORT WALL 2:

Length (open area side) = 5' - 4.5"
= 4.625'

height = 10'

area of wall = 46.25 Sq.ft

area of W10 installed = 3' x 2'
= 6 sq.ft

covered area = 46.25 - 6
= 40.25 sq.ft

(5) LONG WALL 12:

Length = 14' 1.5" = 14.125'
height = 10'

covered area = 141.25 sq.ft

(6) LONG WALL 3:

→ Length (inside open area) = 14.125'
height = 10'

covered area = 141.25 sq ft

Length (other side) = 14.125' - 4.5" - 4.5"
= 13.375'

height = 10'

→ area of wall = 133.75 sq. ft

→ area of two widows = $W_2 + W_8$

= (6.5 x 6.5)' + (3.375 x 2)
= 49 sq. ft

→ area of doors = D_3
= (2.5 x 7)'
= 17.5 sq. ft

covered area = 133.75 - 49 - 17.5
= 67.25 sq. ft

⑦ SHORT WALL 3:

Length = 4' 7.5" = 4.625'
height = 10' = 10'

covered area = 46.25 sq. ft

⑧ SHORT WALL 6 :

Length = 9.5833'
height = 10'

covered area = 95.833 sq. ft

(9) SHORT WALL 5:

→ Length (inside kitchen) = 9'7" = 9.5833'
height = 10'

covered area = 95.833 sq.ft

→ Length (outside Kitchen) = 9'7" - 4.5"
= 9.2083'
height = 10'

covered area = 92.083 sq.ft

(10) SHORT WALL 4:

→ Length = 12.5'
height = 10'

covered area = 125 sq.ft

(11) SHORT WALL 7:

→ Length (facing bedroom) = 4'
→ Height = 10'

covered area = 40 sq.ft

→ Length (facing stairs) = 4'

height = 10'

area of wall = 40 sq. ft

area of door = 3 × 7 = 21 sq. ft

| covered area = 19 sq. ft |

(12) LONG WALL 5:

→ Length (inside kitchen) = 6'-7½"
= 6.625'

height = 10'

| covered area = 66.25 sq. ft |

→ Length / height (facing stairs) = 6'.625'
= 10'

area = 66.25 sq. ft

area of door = 2.5 × 7 = 17.5

| covered area = 66.25 - 17.5 = 48.75 ft² |

(13) LONG WALL 4:

→ Length (facing bedroom) = 6.625'
height = 10

covered areal = 66.25 sq.ft

→ Length (facing Lounge) = 6.625'

covered areal = 66.25 sq.ft

(14) LONG WALL 6:

→ Length (facing stairs) = 3'1½" + 4.5'
= 3.5'

height = 10'

covered areal = 35 sq.ft

→ Length (facing powder room) = 3.125'
height = 10'

covered area = 31.25 sq.ft

(15) SHORT WALL 9:

Length = $11'2'' = 11.1667'$

height = $10'$

covered area = 111.667 sq. ft

(16) SHORT WALL 8:

Length = $13'3'' = 13.25'$

height = $10'$

covered area = 132.5 sq. ft

(17) LONG WALL 7:

→ Length (facing drawing) = 10'3" − 8'
= 2.25'

height = 10

covered area = 22.5 sq.ft

→ Length (facing lounge) = 2.625'

height = 10

covered area = 26.25 sq.ft

(18) LONG WALL 9:

→ Length (facing drawing) = 10.25'
height = 10'

covered area = 102.5 sq.ft

→ length (facing lawn) = 10.25' + 4.5" + 4.5"
= 11'

height = 10'
area of wall = 110 sq.ft
area of window = 8 × 6.5 = 52

covered area = 58 sq. ft

(19) LONG WALL 8:

→ Length (facing porch) = 8'

height = 10'

covered area = 80 sq. ft
: 21

→ Length (facing powder room) = 8' - 4.5"

= 7.625'

height = 10'

area = 76.25 ft²

area of door in wall = 21 + 6 = 27 ft²

covered area = 49.25 ft^2

(20) LONG WALL 11 :

→ Length (facing lawn) = 22.5' - (4.5+2)' - 8'
= 13.75'

height = 6'

covered area = 82.5 sq.ft

→ length (facing street) = 22.5' - 8'
= 14.5'

height = 6'

covered area = 87 sq.ft

(21) SHORT WALL 15 :

→ Length = 6.5'
height = 10'

covered area = 65 sq.ft

(22) SHORT WALL 10:

→ Length (inside powder room) = 6.5'
Height = 10'

covered area = 65 sq.ft

→ Length (facing lounge) = 6.5' + 4.5"
= 6.875'

height = 10

area of wall = 68.75 ft²
area of door = 2 × 7 = 14 ft²

covered area = 68.75 - 14
= 54.75 sq.ft

(23) SHORT WALL 11:

Length = 10'3" = 10.25'
Height = 10'

covered area = 102.5 sq.ft

(24) SHORT WALL 12:

Length = (facing drawing) = 10.25'
Height = 10'

covered area = 102.5 sq.ft

→ Length ~~(facing lounge)~~ (facing porch)

= 10'3" − 9" + 9" + 9"

= 11.75'

height = 10'

area of wall = 117.5 ft²

area by door = 21 ft²

covered area = 96.5 sq.ft

(25) SHORT WALL 14:

length = 6' − 4.5"

= 5.625'

height = 6'

covered area = 33.75 sq.ft

(26) SHORT WALL 13:

length = 15'6" − 6.125' = 9.375'

height = 10'

covered area = 93.75 sq.ft

(27) SHORT WALL (Boundary wall with porch)

Length = 6'
height = 6'

covered area: 36 sq.ft

TOTAL COVERED AREA:

36 + 58

= 3751.836 sq.ft

TOTAL AMOUNT OF DRY PLASTER OF RATIO 1:6

volume of mortar = plaster coved x plaster thickness

$= 3751.836 \times \frac{1}{24}$

$= 156.3265\ ft^3$

(adding 30% to fill upto join in cover surface)

volume of mortar $= 156.3265 + (156.33 \times 0.3)$

$= 203.23$ cft

(increase by 25% for total dry volume)

Dry volume of mortar = 254.036 cft

amount of cement = 63.51 cft

bags of cement $= \frac{63.51}{1.226} = 51.8 \approx 52$ bags

amount of sand = 63.51 x 3

= 190.53 cft

covered area of roof to be plastered = (7.25 x 5) + (14.125 x 4.625)

= 604.854 sqft

volume of plaster = 604.854 x $\frac{1}{24}$

= 25.2 cft

(adding 30% to fill joints & covers)

= 32.763 cft

(adding 25% to get dry mortar)

= 40.954 cft

→ amount of cement = $\frac{40.954}{4}$ = 10.34 cft

→ bags of cement = $\frac{10.34}{1.22}$ = 8.35 bags

amount of sand = 10.34 x 3

= 31.02 cft